Graphical Abstract

Distilling the Complexity of Agent-Based Simulations into Textual Explanations via Large Language Models

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Highlights

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- Three to five bullet points, 85 characters max each including spaces
- Employed optimized prompt engineering techniques for effective report generation.
- Used extractive and abstractive summarization methods to condense explanations.

Distilling the Complexity of Agent-Based Simulations into Textual Explanations via Large Language Models

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Abstract

Explanatory modeling is essential for transforming complex simulation outcomes into understandable narratives that support decision-making, especially among non-experts. This study addresses the challenge of conveying Agent-Based Model (ABM) simulation outcomes in a comprehensive and accessible manner. Given the scalability issues and complexity of ABM data, we propose a novel approach leveraging Large Language Models (LLMs), specifically GPT-4o and Claude 3.5 Sonnet, to generate clear, unbiased reports. Our methodology involves extracting static and dynamic data from NetLogo simulations, generating context and trend analyses through optimized prompt engineering, and summarizing the results using both extractive and abstractive techniques. We evaluate the generated reports through qualitative and quantitative methods, including human assessments and automated metrics. Our findings demonstrate the potential of LLMs in enhancing the interpretability of ABM outputs, thus aiding policymakers in making informed decisions.

Keywords: 1 to 7 keywords, Agent-Based Modeling, Claude, Decision-Making, GPT, Large Language Models, NetLogo, Simulation

1. Introduction

Explanatory modeling plays a crucial role in interpreting complex simulation outcomes, transforming raw data into understandable narratives that

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can inform decision-making. This approach is particularly beneficial for policymakers who may not have the technical expertise to interpret raw simulation data directly and therefore might be less able to put the model's insights into practice [45].

Agent-based modeling (ABM) simulates interactions of agents to assess their effects on the system as a whole. These models are increasingly used and successful across many domains including ecological and socio-ecological systems [34] [22], sociology [36] or political decision making [11].

The challenge lies in conveying the simulation outcomes in a manner that is both comprehensive and accessible. Agent-based models are powerful tools for understanding complex systems, but their results can be challenging to communicate due to the difficulties in analyzing and interpreting the multidimensional output data [14]. This difficulty is exacerbated when trying to communicate findings [24], particularly to non-experts who need to make informed decisions based on these outcomes.[13]

As the complexity of the simulation increases, so does the amount of data generated, making it harder to distill this information into concise, yet comprehensive insights, requiring always more sophisticated analytical approaches [14]. Hence, our goal is to generate clear and concise reports from ABM simulations, at a level of readability and details that would be appropriate for non-modeling experts. [1].

In this paper, we propose a novel approach to generate reports from ABM simulations. Since Large Language Models (LLMs) have demonstrated significant potential in generating explanations and supporting decision-making in complex domains FIND REFERENCES RELATED TO SIMULATION, we employ this technology to generate clear explanations from ABM's output.

The remainder of this paper is organized as follows. In Section 2, we provide background information, including related work on agent-based modeling and explanatory models. In Section 3, we detail our methodology, including the processes of data extraction, report generation, and our evaluation methods, both qualitative and quantitative. Our results are presented in Section 4, followed by a discussion of our findings in Section 5. Finally, we provide concluding remarks in Section 6.

2. Background

2.1. Simulation Outputs of Agent-Based Models

Agent-Based Models (ABMs) generate a significant amount of data by simulating interactions among individual agents within a system. Indeed fields like agent-based traffic simulation are already implementing methods to tackle the large scale simulation and successfully developed solutions used at a production level [46]. Systems like LAMBDA are already demonstrating the ability to manage and analyze large-scale datasets generated by multi-agent systems, effectively addressing data volume challenges in practical applications [29]. These interactions result in complex datasets, necessitating the transformation of raw data into comprehensible formats, typically through visualizations or statistical summaries, to support decision-making by non-experts [14]. This process is essential, as the sheer volume and complexity of the data produced by ABMs can overwhelm users, making it difficult to extract actionable insights directly from the raw outputs [1].

2.2. LLMs and Simulation

Large Language Models (LLMs) have been increasingly used in various simulation tasks, particularly in the development of simulation models using tools such as Netlogo [30]. These models, such as GPT-4 and Claude 3.5 Sonnet, have demonstrated significant potential in generating explanations and supporting decision-making in complex domains. For instance, they facilitate collaboration between computer scientists and domain experts, enhancing the interpretability and utility of simulation outcomes [16]. LLMs have also been utilized in environmental modeling, providing sophisticated analyses and projections [38], and in agent-based simulations to add narrative depth and realism [8].

Recent research by Akhavan and Jalali (2024) provides a systematic approach for integrating generative AI across the simulation research continuum, emphasizing the importance of using these tools to complement rather than replace critical thinking in simulation modeling [28]. Additionally, Ghaffarzadegan et al. (2023) illustrate the potential of Generative Agent-Based Modeling (GABM) in capturing social system dynamics by coupling mechanistic models with LLMs, showcasing how these models can represent human decision-making in social settings [6]. Furthermore, Hosseinichimeh et al. (2024) developed the System Dynamics Bot (SD Bot), which uses LLMs

to automate the creation of causal loop diagrams from textual data, demonstrating how these models can improve the efficiency of system dynamics modeling [5].

Recent research underscores the capability of LLMs to accelerate the development of system dynamics simulations, aiding in tasks such as model creation, error identification, and translation into executable code. For example, GPT-4 has been used to enhance the construction and refinement of system dynamics models, making the process more efficient and reducing human error [51]. These models can process and interpret large datasets, making them suitable for explaining simulation outcomes. Prior research has primarily explored LLMs' capabilities at a conceptual level, highlighting their potential but not fully implementing practical solutions for generating clear and actionable explanations from simulation data [17].

2.3. Explaining and Summarizing in LLMs

The task of explaining complex models using LLMs has shown success across various domains. However, the output from LLMs can often be too verbose for practical use, necessitating summarization. Executive summaries typically require concise and focused content yet powerful for the readers to take action [20], often coming down to 1 or 2 pages with no more than 1000 words [37], and this can be achieved through either extractive or abstractive summarization techniques. Extractive summarization selects key sentences from the text, whereas abstractive summarization generates new sentences that convey the core ideas succinctly. Notable models for these tasks include BERT (Bidirectional Encoder Representations from Transformers) [52] for extractive summarization and BART (Bidirectional and Auto-Regressive Transformers) for abstractive summarization [39]. Both methods aim to distill large volumes of text into more manageable and readable forms, making the information accessible to a broader audience.

2.4. LLMs in Practice : Prompt Engineering and Evaluation

Effective prompt engineering is critical for leveraging the capabilities of LLMs in generating high-quality explanations [27]. Rather than relying on random attempts, prompts must be carefully designed to include essential elements such as roles, examples, and insights. This structured approach ensures that the generated content is relevant and informative [26]. Evaluating the results produced by LLMs involves both quantitative and qualitative

methods. Quantitative analyses often use metrics such as BLEU, METEOR, and ROUGE, which are standard in summarization tasks, to compare the generated content against a ground truth [44, 47]. Additionally, readability can be assessed using the Flesch Readability Score [49]. Qualitative evaluation involves human readers who assess the content based on criteria such as faithfulness and coverage. Given the variability among raters, it is crucial to provide training to ensure consistent and reliable evaluations [12].
By using the Cohen's Kappa, we can determine whether our raters are reaching an agreement or not and thus let them independently rate our reports.

BLEU (Bilingual Evaluation Understudy) measures the precision of n-grams in the generated text compared to a reference text, using a scale from 0 to 1, where higher scores indicate better quality. Prior studies showed performances reaching 24.58% for this metric on summaries from BART [9] and 63% from BERT [4] . METEOR (Metric for Evaluation of Translation with Explicit ORdering) evaluates translation quality based on precision, recall, and alignment, producing scores ranging from 0 to 1, with higher scores representing better quality. Peers have already obtained results for this metrics on text summarized using BART (73.72%) [9] along with BERT (33%) [7]. ROUGE (Recall-Oriented Understudy for Gisting Evaluation) measures recall by comparing the overlap of n-grams between the generated text and reference texts. It includes several variants, such as ROUGE-N (measuring n-gram overlap), ROUGE-L (measuring longest common subsequence), and ROUGE-W (measuring weighted longest common subsequence), with scores typically ranging from 0 to 1, where higher scores indicate better performance. Prior works display results for ROUGE-1 that are approximately of 26.60 for BERT summaries and of 52.53 for BART summaries [3], for ROUGE-2 results approach 11.11 for BERT summaries and 21.83 for BART summaries [3], and lastly for ROUGE-L we have 24.29 for BERT and 49.75 for BART [3]. The Flesch Readability Score assesses the ease of reading a text based on sentence length and word syllable count, with scores ranging from 0 to 100. Higher scores indicate easier readability, with 90-100 being easily understandable by an average 11-year-old student, and 0-30 being best understood by university graduates. Previous studies on LLM have shown a Flesch Reading Score between 24.88 and 58.01 [33]. Cohen's Kappa is a statistical measure used to evaluate inter-rater reliability for qualitative (categorical) items. It accounts for the possibility of the agreement occurring by chance. The Kappa score ranges from -1 to 1, where 1 indicates perfect agree-

ment, 0 indicates no agreement beyond chance, and negative values indicate agreement less than chance. Previous studies claim a kappa between 0.93 [12] and 0.48 [15].

Qualitative assessment frequently relies on human evaluators, although there is ongoing research attempting to extend this to Large Language Models (LLMs) [2]. When dealing with a large volume of outcomes, it often necessitates the involvement of multiple individuals. Previous studies have established a structured process to ensure consistency and reliability [12] : initially, a first batch of outcomes is collaboratively rated to establish a standard. Subsequently, another batch is independently rated by pairs of raters to measure interrater agreement using Cohen's Kappa. Finally, the remaining results are individually rated, assuming a satisfactory level of interrater agreement has been achieved. A common challenge encountered is the high variability in Cohen's Kappa values, indicating inconsistent agreement among raters. To address this, researchers are increasingly defining detailed rubrics to enhance precision and clarity in ratings. Two predominant approaches are often employed: descriptive rubrics [32], which provide detailed criteria for each level of performance, and holistic rubrics [12], which offer a single overall score based on a general impression of performance. These rubrics aim to standardize the evaluation process, reduce variability, and ensure more reliable and consistent assessments across different raters.

3. Methods

3.1. Overview

Our methods are divided into three main parts (Figure 1). First, the Agent-Based Model is specified in NetLogo (subsection 3.2). Given a set of parameters, we run several simulations to obtain results from the model. We also extract data from the NetLogo file: we parse the documentation, the Graphical User Interface (GUI), and the comments from the code. Then we generate a report using this information (subsection 3.3). GPT-4o and Claude 3.5 Sonnet are used to generate both a context and a trend analysis, then a summarization algorithm reduces the number of paragraphs to the length suitable for a report. We optimize the prompt engineering process using a systematic design of experiments to ensure the generated reports are both accurate and informative. Finally we evaluate the report (subsection

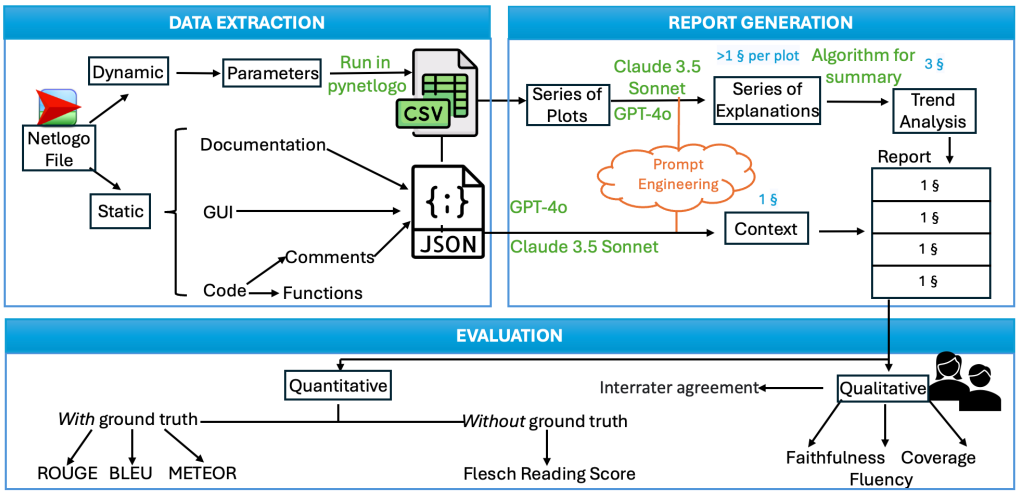

Figure 1: Our procedure consists in three main parts. Each part is explained in turn, in a dedicated subsection

Parts	Use	Python Library
Data Extraction	Running the simulation	pynetlogo (0.5.2)
	Extracting the static data	re (2.2.1), json (2.0.9)
Report Generation	Plotting the data	pandas (1.5.3), matplotlib (3.8.3)
	Generating explanations	openai (1.14.3), anthropic (0.29.0)
	Converting image to base64	base64 (python 3.10.9)
	Combining prompt features	itertools (python 3.10.9)
	Summarizing	transformers (4.41.2)
Evaluation	Scoring	nltk (3.7), rouge_score (0.1.2)

Table 1: Python Libraries used including their role in our method

3.4) by using a qualitative method, and a quantitative method relying on a ground truth for certain metrics.

3.2. Data Extraction

Data extraction involved processing NetLogo simulation files. Our research uses three well documented social agent based models from the web-

site Comses.net. These files were categorized into dynamic and static components.

The dynamic data consisted in parameters we could modify. Thanks to this part of the file, we were able to run the simulations and retrieve the data using pynetlogo (Table 3.1). Therefore, we have set the initial parameters, the metrics to follow throughout the experiment, the halting condition and ran the experiment. For example a parameter could be $numberof\text{-}agents = 1,000$, a metric could be "mean [incum-quantity] of turtles" and a halting condition could be $max_ticks = 73,000$. This would mean that the experiment starts with $1,000$ agents, for a maximum of $73,000$ ticks during which we will measure the mean quantity of whole milk consumed by agents. All the results were then exported into CSV format for further analysis.

From the static data, we extracted the documentation, the GUI (the modifiable elements and their range of values) and the comments from the code using regular expressions (Table 3.1). While parsing the documentation, we made sure to not include the default content from Netlogo files that were left unfilled and we also made sure to remove urls as it was subject to hallucinations from the LLMs. From the GUI, we have made sure to only pick relevant information (i.e the minimum and maximum value as well as the value set for the experiment). This content was saved into JSON files and plays an essential role in giving context to the simulation for the following explanations.

3.3. Report Generation

The report was designed to synthetize the extracted data into meaningful explanations. We specifically used GPT-4o and Claude 3.5 Sonnet as they are both ranked 1st and 2nd in the ChatBot Arena [50] (rank on 1st July 2024).

The first step was to give context to the simulation. Since the JSON files provide the extracted context (Section 3.2), we appended this information to the prompt given to the LLM (Prompt 3.3). We specified to the LLM that the generated paragraph should include the context of the simulation, its goals and the key parameters set for the experiment in a way for the reader to catch the experiment conditions.

The second step was to describe the data from the simulation. To do so, we used the CSV file to plot the different metrics. With the help of GPT-4o and Claude 3.5 Sonnet's vision capabilities, we then gave these plots to the LLM and generated explanations. We optimized the prompt engineering by using a systematic design of experiments known as 2^k factorial design, where k is the number of features considered in the prompt (Prompt 3.3). That is, we automatically generate a series of prompts and record their effect in order to (i) find the best prompt and (ii) assess which parts of a prompt contribute to its effectiveness. Here, we used $k = 3$ features:

- use of a role (also named a persona) [26] [42] (example : "You are an expert in Consumer Behavior")
- an example of expected style, as we already showed that examples in prompts can provide better outcomes in several fields, like causal maps [40] (for instance : " The total number of earthquakes recorded in the region starts at $5,000$ then it declines rapidly over the first 400 time steps. This initial decline could be attributed to the depletion of immediate stress points within the fault lines, as the most susceptible areas release their pent-up energy early in the simulation. It increases briefly at around 500 steps, likely due to the aftershocks and secondary stress points being triggered as the system seeks a new equilibrium. The simulation ends with the number of earthquakes near zero, indicating that the majority of the stress has been released and the system has stabilized.

  The total seismic activity follows the same pattern, starting at $10,000$ events and declining steadily. This steady decline reflects a systematic reduction in seismic energy over time, as the energy distribution within the tectonic plates becomes more uniform. There is low variance across simulation runs in the first 100 steps, but deviations become noticeable afterward. This suggests that while the initial reactions to stress are highly predictable, as time progresses, the system's complexity introduces more variability. This variability could be due to differences in secondary stress points and the non-linear nature of seismic energy release over time.")
- "insights" that might need a reference("When summarizing trends, provide brief insights about their implications for decision makers.").

Prompt 1 : `text` is a feature from the design of experiment. *text* is a part of the prompt changing according to the model and the experiment. Complete prompts can be found in the supplementary material file.

`Role.`

Your goal is to explain an agent-based model. Your explanation must include the context of the model, its goals, and key parameters. For each parameter, include the range of values (if provided) and the value that we have set. Do not write any summary or conclusion.

Parameters

The context and goals of the model are as follows. *documentation*

Once the context and the trend analysis were generated, we needed to remove some unnecessary symbols for the report to only contain relevant information. That's why we deleted all the "#" and " * " symbols found in the API response due to the Markdown format it uses.

In a way to gather all this data into a report, we then used an algorithm to summarize these explanations into 4 paragraphs. For this we distinguished two types of algorithms either extractive or abstractive.

Extractive summarization techniques function by identifying and extracting salient sentences from a source document to generate a concise representation. This approach prioritizes factual content and offers computational efficiency, making it well-suited for summarizing informative texts. In our case we leveraged the BERT [35](Bidirectional Encoder Representations from Transformers) algorithm.

In contrast to extractive summarization, abstractive summarization techniques aim to capture the essential meaning of a source document and produce a concise representation using novel phrasings and potentially new sentences. As part of this type of algorithms, we chose to leverage BART [39] (Bidirectional and Autoregressive Transformers for Pre-training).

Both these methods were giving us a summmary for the context and one for every plot from the trend analysis (i.e between 10 and 14 summaries for the trend analysis). To finally get an organized report we had to gather these summaries into paragraphs. This was done by making sure that the context summary would be a paragraph on its own, and that the following summaries would be equally distributed within 3 paragraphs.

Additionally, the summaries from BERT would not always contain a upper case letter at the beginning of each sentence, which then required a post processing algorithm to go through these summaries and replace the lower case letters by upper case ones. Moreover, we found out that some text within the CSV would not be displayed properly on certain devices, which is the reason we ultimately give the CSV as an input to an algorithm removing all the non-unicode characters. Some other post processing modifications were required on the summaries, especially on the one from BERT concerning punctuation, spaces in parenthesis, or hyphens.

The final report is the concatenation of the context followed by the data analysis paragraphs, for a total length of 4 paragraphs.

Prompt 2 : `text` is a feature from the design of experiment. *text* is a part of the prompt changing according to the model and the experiment. Complete prompts can be found in the supplementary material file.

`Role.`

We have a plot from repeated simulations of an agent based model. Your goal is to describe the trends in details from the plot, mentioning key time steps and values taken by the metric in the plot, and interpreting them based on the context of the model. The report must objectively describe the trends in the data without addressing the quality of the simulation. Do not refer to the plot or any visual in your description. If a plot has very simple dynamics, simply state them without expanding.

`Example.`

`Insights.`

The attachment is the plot representing the average weekly consumption of whole milk per agent

3.4. Evaluation

Our evaluation method is divided into two procedures.

First, using a well known method, human readers perform a qualitative evaluation by ranking the generated reports according to two different criteria (faithfulness, coverage) using the 5-point Likert scale [18] [31] [25]. Since LLM generations often involve a large amount of output, it is common to manually assess only a random sample of these results. Indeed such sampling techniques are used in several fields when it comes to evaluation, like the performance of LLMs in machine translation, [19], evaluation of LLMs capabilities [21] or even the use of GPT to answer questions about patient data from electronic health records [23]. That is why we asked our raters to independently assess a random sample of INSERT NUMBER reports out of 96. We first taught our raters how to assess reports according to faithfulness and coverage on an unrelated model from our study. We then asked them to independently assess 6 examples from our models and ultimately computed their interrater agreement using the Cohen's Kappa metric [41]. The results

were :

$$\kappa_{\text{Coverage}} = 1$$

$$\kappa_{\text{Faithfulness}} = 0.333$$

After we supervised a meeting with the raters for them to justify their assessment and come to an agreement on a number for the cases they rated differently, several general observations were made. There were discrepancies in the interpretation of metrics, with confusion over specific values and steps, and a lack of precision in describing trends. The raters often disagreed on the scoring, but occasionally reached consensus after discussion.
TO BE CONTINUED

The second part of our evaluation is an automated quantitative process, consisting of two parts. A first set of metrics is using a ground truth reference we wrote based on the simulation data, representing a report we would expect in the best case. The metrics using this reference are BLEU [43] [44] [48], METEOR [48], ROUGE-1, ROUGE-2, ROUGE-L [43] [47] [48]. The second part of our quantitative method does not use a reference; rather, it focuses on the readability of the report and thus evaluates it based on the Flesch Readability Score [49].

4. Results

4.1. Quantitative Results

4.1.1. Performance of Language Models

We evaluated the performance of two leading language models, GPT-4o and Claude 3.5 Sonnet, in generating comprehensive and objective explanations from Agent-Based Model (ABM) outputs. The automated metrics used for evaluation include BLEU, METEOR, ROUGE-1, ROUGE-2, ROUGE-L, and Flesch Reading Ease scores. These metrics provide insights into the quality, accuracy, and readability of the generated texts.

The GPT model consistently outperformed the Claude model across all quality and accuracy metrics (Table 4.1.1) (BLEU, METEOR, ROUGE-1, ROUGE-2, ROUGE-L). However, the text generated by Claude was found

LLM	BLEU	MET	R-1	R-2	R-L	FRE
Claude	0.04259	0.23371	0.44485	0.12047	0.18548	51.28
GPT	0.04853	0.24594	0.45972	0.12972	0.18904	46.73

Table 2: Comparison of Automated Metrics for Different Language Models

to be slightly more readable according to the Flesch Reading Ease score. This indicates that while GPT provides more accurate and comprehensive explanations, Claude may be preferred for simpler, more accessible texts.

4.1.2. Impact of Experimental Design Factors

To further understand the influence of different prompt engineering strategies, we evaluated the combined impact of three factors: the use of a role (persona), the inclusion of an example of expected style, and the provision of insights. The results are summarized in the following table.

Role	Ex	Ins	BLEU	MET	R-1	R-2	R-L	FRE
No	No	No	0.04601	0.23648	0.44899	0.12390	0.18723	49.26
No	No	Yes	0.04351	0.23871	0.45070	0.12293	0.18674	46.86
No	Yes	No	0.04761	0.24094	0.45388	0.12727	0.18778	51.16
No	Yes	Yes	0.04511	0.24317	0.45558	0.12630	0.18729	48.75
Yes	No	No	0.04601	0.23648	0.44899	0.12390	0.18723	49.26
Yes	No	Yes	0.04351	0.23871	0.45070	0.12293	0.18674	46.86
Yes	Yes	No	0.04761	0.24094	0.45388	0.12727	0.18778	51.16
Yes	Yes	Yes	0.04511	0.24317	0.45558	0.12630	0.18729	48.75

Table 3: Comparison of Automated Metrics by Combined Design of Experiment Factors

The results indicate that including examples and insights generally improves the performance of the language models across most metrics (Table 4.1.2). The role (persona) feature had a less significant impact, suggesting that the clarity and specificity provided by examples and actionable insights are more critical in generating high-quality explanations.

4.2. Qualitative Results

5. Discussion

5.1. Challenges and Future Directions

While the integration of LLMs with simulation outputs shows great promise, there are significant challenges to address. These include improving the ac-

curacy and relevance of explanations, handling the variability in input data, and ensuring the interpretability of model outputs while handling LLM's hallucinations. Future research should focus on developing robust frameworks for integrating LLMs with simulation models, enhancing prompt engineering techniques, and establishing standardized evaluation metrics to consistently measure the performance of LLMs in generating explanations. Furthermore, leveraging advances in retrieval-augmented generation and fine-tuning methods could enhance the applicability of LLMs in diverse simulation contexts.

6. Conclusion

7. Glossary

ABM: Agent-Based Modeling, a discrete simulation technique where individual entities (agents) interact with each other and with their environment.

NetLogo: A simulation environment for agent-based modeling.

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